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How does customer enterprise digitalization improve the green total factor productivity of state-owned suppliers: From the supply chain perspective

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ABSTRACT

Environmental problems arising from global industrialization highlight the urgent need to green supply chains. However, the intensification of digitalization and global supply chain risks underscore the leading role of state-owned suppliers. From the supply chain perspective, this study analyzes the impact of the digitalization of downstream customer enterprises on the green total factor productivity (GTFP) of state-owned upstream suppliers using data on Chinese listed companies from 2007 to 2022. The benchmark regression shows that the digitalization of downstream customer enterprises significantly improves the GTFP of state-owned upstream suppliers. Mechanism testing shows that the digitalization of downstream customer enterprises promotes the GTFP of state-owned suppliers by reducing implicit costs and increasing the proportion of highly educated human resources. In improving the GTFP of state-owned suppliers, the external contractual environment, internal environmental responsibility, and market competitiveness act as "accelerators," whereas the intensity of environmental regulation acts as a "speed bump" in the short term. In contrast to existing research, which mainly focuses on the impact of enterprise digitalization on the production efficiency of internal or upstream suppliers, this study facilitates the understanding of the pillar role of state-owned suppliers in the national economy and their stabilizing function in the supply chain.

1. Introduction

With the advancement of global industrialization, the deterioration of the ecological environment has become a worldwide problem, making green supply chains a trend and an inevitable choice [28]. With the rapid development of information technology and the promotion of global trade integration, the main bodies of the value chain are closely related, and communication is highly efficient, thus providing new opportunities for a green supply chain [7]. However, owing to natural disasters [19], political conflicts [1], policy uncertainty [17], and other factors, the supply chain ripple effect frequently occurs with significant negative effects [25]. Compared with private enterprises that pursue value maximization, state-owned enterprises shoulder the burden of social stability and development and play an important role in stabilizing the supply chain and global economy. Additionally, the green transformation of state-owned enterprises can drive the transformation and upgrading of upstream and downstream enterprises through demonstration effects [52]. Therefore, within the supply chain, state-owned suppliers not only play a pivotal role in managing supply chain costs and minimizing disruptions but also enhance the overall green efficiency of downstream enterprises and the supply chain as a whole [18]. Under the pressure of sustainable development in the supply chain, state-owned suppliers are advised to connect with customer enterprises with high levels of digitalization to maintain their stabilizing function. Moreover, the impact of the upward transmission effect of customer enterprises' digital transformation on the green total factor productivity (GTFP) of state-owned supplier enterprises must be studied.

Affected by economies of scale and specialized division of labor, state-owned suppliers possess business knowledge structure, organizational capacity, and business logic that primarily revolve around traditional business architecture. They also lack open awareness of resource

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co-construction and sharing, cross-department collaboration, and so on. Consequently, achieving breakthroughs in new business models is difficult for these suppliers owing to their insufficient experience in digital business operations [43]. However, stakeholders' increasing expectations have forced state-owned suppliers to reform their corporate governance systems and improve their operations to reduce deficits and promote low-carbon transformation, thereby establishing them as strategic tools to secure national competitiveness and promote sustainable development [30]. In addition, state-owned suppliers' green development is subject to external shocks. For example, driven by Industry 4.0, customer enterprises shift their supply chain relationships from the traditional supply-oriented to demand-oriented owing to their green and digital business needs, thus prompting state-owned suppliers to improve their production methods, engage in low-carbon business transformation, and ultimately improve GTFP [22].

Therefore, in developing the national economy, upstream state-owned suppliers can promote the stability of the industrial supply chain and the coordinated development of large, medium, and small enterprises. They also act as stabilizers. Their specific functions are as follows: (1) State-owned suppliers undertake long-term, high-investment, and high-risk projects (e.g., transportation, energy, information, and other infrastructure construction), maximize resource flow, and optimize resource allocation under government intervention, ultimately reducing market circulation costs [14]. (2) State-owned suppliers fulfill their social responsibilities by absorbing surplus labor, controlling market prices, strengthening quality monitoring from the source of manufacturing, and improving the level of low-carbon development. The objective is to develop a mutually beneficial economic environment. (3) In the event of emergencies or disasters, state-owned suppliers can take counter-cyclical measurements to quickly respond to market crises and relieve downstream capital pressure and offset market failure by providing credit support and delayed payment for customer enterprises. Furthermore, the transformational actions of enterprises within the supply chain can influence the sustainable development of domestic upstream and downstream enterprises and the supply chain itself, as well as the sustainable development of enterprises in other countries through the global supply chain. With China being a significant player in the global supply chain, conducting research on the impact of the digitalization of Chinese customer enterprises on the GTFP of state-owned suppliers has a certain reference value for other countries.

Considering the integration of digitalization and greening, this study explores the impact of the digitalization of downstream enterprises on the GTFP of state-owned upstream suppliers. The contributions of this study are as follows:

(1) Existing studies have overlooked the sustainable development needs of the global supply chain and the stabilizing role of state-owned suppliers. In addition, enterprises' green development is influenced not only by internal factors but also by external shocks. Therefore, considering the factors influencing supply chain configuration, this study extends the relationship between digitalization and enterprises' GTFP to the supply chain level, examines how transformation within customer enterprises radiates to suppliers' production processes, and verifies the promoting effect of digital technology on the real economy.

(2) Existing research has analyzed the impact mechanisms of ripple, backwardness, and spillover effects on the total factor productivity of supply chain entities. However, in the context of digitalization and intensified global risks, the global supply chain has gradually shifted from being supply-oriented to being demand-oriented. "Demand-led innovation" has become a driving force of technological progress. Therefore, conducting research from a supply-driven perspective does not meet the requirements of the new situation. Given the market's demand for greening and the upward transmission effect of enterprise digitalization, this study examines the effect of the digitalization of downstream customer enterprises on the GTFP of state-owned upstream suppliers. This study also supplements the impact mechanism of the vertical contagion effect and provides a new perspective for examining the relationship between state-owned suppliers and their customers.

(3) Existing studies have ignored the capability of customer enterprises to adjust the GTFP of state-owned suppliers through digitalization in the context of the change in the dominant power of the global supply chain. Under the goal of sustainable development, this study considers not only state-owned suppliers as important "stabilizers" but also the incomplete contract and information transmission theory by using data on China's listed enterprises in 2007–2022. From the supply chain perspective, this study demonstrates the moderating role of customer enterprise digitalization in the impact of the external policy environment and the influence of the internal management factors of state-owned suppliers on their GTFP.

The remainder of this paper is organized as follows. Section 2 presents the review of related literature, and Section 3 proposes the research hypothesis. Section 4 describes the research design, including the data sources and model settings. Section 5 discusses the empirical results, robustness testing, and mechanism analysis. Section 6 presents the moderating effects of the external and internal factors. Section 7 summarizes the conclusions and policy recommendations.

2. Literature review

Digitalization is a high-level transformation based on digitization, breaking through the constraints of traditional element boundaries and achieving deep integration of all-round elements with digital technology [51]. Three types of measurement methods exist for enterprise digitalization. (1) Scale method. The scale method measures the degree of digitalization of an enterprise from multiple dimensions on the basis of surveys of enterprise managers, employees, and cooperative enterprises. However, this method is vulnerable to the respondents' subjective cognition. (2) Index method. This method indirectly calculates the degree of digitalization of enterprises through financial statements but has evident limitations in data collection and estimation. (3) Text analysis. This method uses virtual variables to measure whether the enterprise is undergoing a transition of digitalization, but it cannot reflect the "intensity" of enterprise digitalization. Overall, the basic idea for measuring digitalization is to construct an "enterprise digitalization" feature thesaurus using technologies such as big data, artificial intelligence, blockchain, and cloud computing. Subsequently, the annual report extraction results of listed enterprises are matched with the "enterprise digitalization" feature thesaurus, and the frequency of words is then used to describe the degree of enterprise digitalization [40].

The GTFP reflects the environmental pollution and resource consumption caused by profitable behavior and can thus be used to comprehensively evaluate the production efficiency of enterprises [49]. The GTFP can be measured using two methods: (1) parameter estimation method, which calculates the GTFP of enterprises by setting a function that includes various production factors, such as energy consumption and pollution emissions; and (2) non-parametric methods, which measure the GTFP according to the distance between the decision unit and the stochastic frontier by constructing a stochastic frontier of the production function [12]. Data envelopment analysis (DEA) is a widely used method for flexibly handling multi-input or multi-output problems without the need for preset functional relationships between inputs and outputs. When applying the DEA method, one approach is to consider pollutants as cost inputs; however, the resulting model does not reflect the actual production process and treats pollutants as unexpected outputs. In particular, the super-efficient slacks-based measure (SBM) model proposed by Tone [38] addresses the errors caused by the radial selection of traditional DEA models and further distinguishes

decision-making units with efficiency values equal to 1. Given that the green development of enterprises is a long-term process, the global Malmquist–Lunberger method is usually adopted for vertical comparisons [20].

2.1. Impact of enterprise digitalization on GTFP

Digitalization has changed the business logic, value propositions, external interactions, and link channels of enterprises. It has also improved internal and external information transmission efficiency and enhanced green production and management capabilities [5,50]. Specifically, digitalization affects enterprises' GTFP in the following ways:

(1) Digitalization improves enterprises' information acquisition and communication abilities and reduces external transaction costs. The application of digital technology can significantly reduce the costs of information retrieval, transmission, and application in enterprises. In particular, digital economy platforms have reshaped how companies organize production and innovation, optimized the external information environment, and reduced the costs of external coordination arising from information barriers. Real-time tracking of transaction data not only reduces losses caused by human error but also continuously optimizes transaction strategies and reduces regulatory costs. Owing to the noncompetitive nature of data, enterprises can use their own data and share data and technology from other enterprises, thereby generating knowledge spillover and knowledge creation effects through learning and imitation to improve their GTFP [34].
(2) In the context of digitalization, enterprises reconstruct traditional production and operation modes through information technology, providing an impetus for innovation enhancement at the "input–output" level. Through the application of technologies such as big data and artificial intelligence, enterprises can interlock digital technologies and traditional production modes, including reorganizing and optimizing production resources. They can also build an entire industrial chain of production, transportation, consumption, and recycling. Furthermore, enterprises can improve resource utilization efficiency and reduce pollution emissions by enhancing the specialized division of labor to improve the quality and efficiency of the industry [10]. Additionally, the development of the digital economy is conducive to the birth of new forms, modes, and industries, creating new growth points for economic development. Such development is instrumental in constructing digital management and sales modes, improving resource efficiency, promoting technological innovation, and empowering the economic market to develop toward decarburization [2].
(3) As a new production factor brought about by digitalization, data capital, combined with information and communication technology, can directly improve green production efficiency and complement enterprise management. Data capital can penetrate the production, management, decision-making, and core business of enterprises. Meanwhile, enterprises can reduce resource mismatch and directly improve their GTFP by analyzing and forecasting based on historical data such as customer information, sales records, and production processes. Moreover, they can use data to improve the efficiency of information processing, transmission, and communication, which in turn reduces transaction costs and facilitates the accumulation of innovation potential; hence, enterprises gain the necessary information efficiency and receive growth opportunities for market trading behavior [9,44].

2.2. Impact of digitalization of customer enterprises on state-owned suppliers

State-owned suppliers are assigned greater responsibility for sustainable development. On the one hand, the attributes of state-owned suppliers require them to actively undertake social responsibility, employ green practices, reduce resource consumption and pollution emissions, and implement green innovation [46]. On the other hand, state-owned suppliers, under environmental regulations and pressure from public opinion, should enhance their competitive advantage by improving their green production capacity [35]. Digital technology has improved enterprises' ability to share information. Customer enterprises collaborate with their state-owned suppliers to promote green supply chain construction; improve the green efficiency of the supply chain and the whole system of "procurement, production, transportation, and sales"; and provide opportunities for state-owned suppliers to improve their GTFP.

Specifically, the digitalization of enterprises improves the degree of upstream and downstream information sharing and reduces the cost of obtaining external information for state-owned suppliers. It also facilitates the spillover of advanced knowledge and technology and creates learning opportunities for state-owned suppliers. Meanwhile, a network information system improves the ability of customer enterprises to process massive amounts of information. This case is conducive to outputting additional effective information, improving the information transparency of the supply chain, reducing the cost of information search and verification for state-owned suppliers, and compensating for the efficiency loss caused by the low efficiency of internal information transmission of state-owned suppliers. Additionally, the universality and accessibility of digital technology can effectively improve the speed of technology diffusion [23]. This case is conducive to the spillover of customer enterprise knowledge and technology along the supply chain [41]. The state-owned suppliers in the upstream segment of the supply chain can learn and imitate the green and innovative technology and production mode of customers' enterprises through "learning by doing." They can also produce a collaborative innovation effect and improve state-owned suppliers' green production performance. In addition, they have great potential to promote commerce, product, and service innovation [42].

Overall, digitalization has improved the efficiency of information transmission and has had a positive impact on alleviating obstacles to information exchange and enhancing market response speed. However, constrained by macro management factors, state-owned suppliers engage in relatively conservative decision-making, which results in poor market flexibility and sensitivity. Nevertheless, the digitalization of enterprises has broken down information barriers among different links. State-owned suppliers are thus able to explore and grasp customer needs, accurately provide products and services, and provide auxiliary decision-making support [36]. Information technology and big data analysis methods brought about by digitalization have improved the quality and efficiency of information transmission in capital markets. These methods have also enhanced the matching, timeliness, and adaptability of upstream suppliers' responses to changes in downstream demand. Furthermore, they have reduced the possibility of decision-making errors and operational risks for state-owned suppliers due to information asymmetry [33]. Additionally, state-owned suppliers' targeted adjustments and adaptations according to market demand are conducive to clarifying their market positioning, concentrating resources to form core competitiveness, and seeking new growth points [24].

Existing research has been conducted from the perspectives of digitalization, supply chain relationships, efficiency improvement of state-owned enterprises, and sustainable development. The related research can be divided into two categories. The first category focuses on the analysis of economic consequences, exploring the impact of digitalization and its chain reaction on promoting upstream and downstream

technology transfer and experience sharing, improving the supply chain response rate and intensive costs, and promoting the overall performance of the supply chain [29]. The second category focuses on the study of noneconomic consequences and discusses the potential of monitoring technology and intelligent management systems brought about by digitalization in energy consumption, green innovation, energy conservation, and emission reduction [21,31]. However, these studies did not open the "black box" of digitalization affecting the sustainable development of the global supply chain and ignored the stabilizing role of state-owned suppliers.

3. Research hypothesis

3.1. Impact of digitalization of customer enterprises on GTFP of state-owned suppliers

With the shortening of the product life cycles and refinement of market divisions, upstream and downstream enterprises have become closely related. Moreover, cooperation in production, sales, and other aspects has been strengthened. Accordingly, supply chain relationships have become the main source of competitive advantages for enterprises amid fierce market competition.

On the one hand, information technology is conducive to improving the information processing capacity and dissemination speed of enterprises. Customer enterprises can transmit the demand information accurately and reduce the cost of supply and demand coordination of their state-owned suppliers. Customer enterprises disclose their credit, price, environment, and other information through digitalization. In addition, they improve the information environment among enterprises and reduce the information costs and matching difficulties for both parties. They also have a positive spillover effect on the green production decisions and resource allocation of state-owned suppliers [15].

On the other hand, state-owned suppliers may revise their production plans and accelerate digitalization to stabilize customers. The digitalization of customer enterprises has generated additional green and innovative needs, forcing state-owned suppliers to upgrade equipment and technology. The knowledge and technology spillover of customer enterprises not only reduces the knowledge search cost of state-owned suppliers but also brings them learning and imitation opportunities. This spillover has a positive impact on state-owned suppliers' breaking of path dependence and reconstruction of production processes and management modes [8]. In addition, the environmental pressure of customers caused by market green demand spreads upstream, driving state-owned suppliers to adopt high-quality green innovation strategies to obtain "strategic legitimacy." This case is conducive to improving their green performance [13].

H1. The digitalization of customer enterprises improves the GTFP of state-owned suppliers.

3.2. Effect mechanism of digitalization of customer enterprises on GTFP of state-owned suppliers

Digital technology has the attribute of "capital," which can reduce investment and development costs, and the attribute of "labor." While generating substitution effects in labor supply, digital technology stimulates the generation of new jobs that meet production needs [29]. Specifically, digitalization reduces the degree of information asymmetry in the supply chain and decreases decision-making errors and operational failures caused by information asymmetry. Under the market mechanism, suppliers with information inferiority may compensate for the profit loss caused by incomplete information. In other words, they increase financial leverage and reduce corporate social responsibility investment, which are against their green development [4]. Digitalization enables state-owned suppliers to obtain additional effective information at a low cost, accurately depict customer enterprise portraits, reduce cost stickiness and transaction risks, and acquire further resources available to practice social responsibility. This case is conducive to improving green production efficiency. In addition, the digitalization of customer enterprises has generated new demands and information advantages. State-owned suppliers usually choose to maintain strategic alignments (or alliances) with customers to stabilize supply chain relationships [52]. From the information transmission perspective, customer enterprises may proactively disclose pollution information and encourage suppliers to fulfill their social responsibility to avoid negative media publicity. This even has a positive impact on the social reputation capital and green performance improvement of state-owned suppliers.

However, the impact of digitalization on state-owned suppliers is mainly reflected in cost control and human resource management. According to the green innovation requirements of customer enterprises, state-owned suppliers' production mode adjustment is essentially a skill-oriented technological progress [47]. This case leads to a major adjustment in the demand structure of state-owned suppliers' employees. Highly educated talents with innovative and green knowledge and technological reserves are assigned important responsibilities as support factors. State-owned enterprises need to reform their organizational structure to achieve a professional division of labor and comply with the adjustment of business activities. The application of digital technology has also increased the preference for highly educated employees and demand for high-end talents. This adjustment increases operating costs and high-tech capital investments. Nevertheless, highly educated employees and their specialized knowledge have a positive impact on improving business processes and integrating various aspects of green production and operations. This case is beneficial for improving the efficiency of green production and the operation of state-owned suppliers [3].

H2a. The digitalization of customer enterprises significantly improves the GTFP of state-owned suppliers by affecting their implicit costs.

H2b. The digitalization of customer enterprises significantly improves the GTFP of state-owned suppliers by affecting the proportion of highly educated human resources of state-owned suppliers.

3.3. Moderating effect of external policy environment and internal management factors

At the micro level, the digitalization of customer enterprises can affect the economic and noneconomic performance of state-owned suppliers [16]. The GTFP of state-owned suppliers is mainly affected by the following two aspects: (1) enterprise internal management factors such as enterprise profitability, research and development (R&D) investment, and environmental awareness [37]; and (2) external policy environments such as environmental rules and institutional innovation [47]. Therefore, starting with the factors affecting the economic and noneconomic performance of enterprises, this study considers the two dimensions of external policy environment and internal management factors.

3.3.1. Moderating effect of factors related to enterprises' economic performance

The development of information technology has promoted the vertical integration of the industrial chain, which helps establish information sharing mechanisms among enterprises. Nevertheless, the supplier and demander of products or business activities are mutually independent, leading to cooperative relationships, including competitive and squeezing relationships in vertical supply chains [48].

From the perspective of the external business environment, an information gap exists between suppliers and customers owing to incomplete information and contracts. Moreover, both parties may engage in opportunistic behavior of concealing information. Enterprises with information advantages in the supply chain tend to seize value and

exploit vulnerable parties in interest games. Particularly, state-owned suppliers with vertical technological cooperation not only face performance decline caused by sunk costs but also bear economic losses, such as shutdown and looking for new partners. A stable trust relationship is conducive to reducing transaction costs for supply and demand sides, enabling the accurate disclosure of information by customer enterprises. This relationship is also conducive to reducing information redundancy, thereby improving the "green bullwhip effect" in the supply chain and the contractual environment for state-owned suppliers. State-owned suppliers gain additional Solow residual and are willing to increase technological investment to improve their specialized production capacity. However, this case may also have a "crowding out effect" on green R&D investment, leading to improved production performance of state-owned suppliers while environmental performance remains unchanged or even declines [6].

From the perspective of internal comprehensive ability, a lack of market competitiveness leads to a decline in enterprises' anti-risk ability. In addition, state-owned suppliers may encounter demand shocks and weakened bargaining power [48].

When confronted with a complex market environment and economic fluctuations, state-owned suppliers must cope with the risk of sudden supply chain suspensions or disruptions. State-owned suppliers with low market competitiveness are unable to mitigate risks through customer diversification and can only accept unfair pricing to stabilize customers, as reflected in a decline in enterprise profitability. In addition, the higher the dependence of state-owned suppliers on customers, the greater the pressure of market competition, and the more unfavorable it is for enterprises to improve their GTFP.

H3a. Improvement in the external contract environment magnifies the promotional effect of the digitalization of customer enterprises on state-owned suppliers' GTFP.

H3b. Market competitiveness strengthens the positive role of customer enterprises' digitalization in improving state-owned suppliers' GTFP.

3.3.2. *Moderating effect of factors related to enterprises' noneconomic performance*

According to system theory, the main power sources for enterprises to implement green decision-making and green production mode are customer pressure and environmental regulation pressure, with the former being essentially derived from the latter [39].

From the perspective of the "passivity" of external regulatory policies, the government formulates an environmental regulatory system to constrain enterprise pollution emissions [45]. Hence, state-owned suppliers need to pay for governance or emissions, resulting in additional environmental regulation "compliance costs." Some enterprises even need to slow down output to temporarily meet environmental regulation requirements. In the short term, pollution control at the expense of production efficiency also produces a "crowding out effect" on state-owned suppliers' for-profit investment, which is not conducive to improving green performance [11].

From the perspective of "initiative" in internal management decisions, state-owned suppliers actively fulfill their social responsibilities and adjust their production models to meet the requirements of "greening" so as to maintain their good image and stabilize supply chain relations. Therefore, under environmental regulations, state-owned suppliers can gain backward spillover effects from the digitalization of customer enterprises. They may also gain differentiated competitive advantages owing to green production modes and innovative products and even trigger "innovation compensation" effects to improve their green performance [47].

H3c. High-intensity environmental regulations weaken the positive impact of the digitalization of customer enterprises on improving state-owned suppliers' GTFP.

H3d. Enterprises' environmental responsibility enhances the positive impact of the digitalization of customer enterprises on improving state-owned suppliers' GTFP.

4. Research design

4.1. Model

This study introduces an econometric model to examine the impact of customer enterprise digitalization on the GTFP of state-owned suppliers. The benchmark econometric regression model is as follows:

$$\ln GTFP_{i,t} = \beta_0 + \beta_1 \ln Cdigi_{i,t} + \sum Controls_{i,t} + \{F\} + \varepsilon_{i,t} \quad (1)$$

where $\ln GTFP_{i,t}$denotes the natural logarithm of the GTFP of state-owned supplier i in year t. The core explanatory variable of this study is $\ln Cdigi_{i,t}$, which indicates the natural logarithm of the weighted average digitalization degree of listed customers among the top five customers of state-owned supplier i in year t. $\sum Controls_{i,t}$ refers to a set of control variables that include year, industry, and region fixed effects. Herein, the benchmark model controls for year t, industry j, and region k, denoted as $\{F\} = \{\varphi_t, \zeta_j, \gamma_k\}$; $\varepsilon_{i,t}$is a random error.

4.2. Core variable

(1) Explanatory variable: customer enterprise digitalization ($\ln Cdigi$)

Enterprise digitalization refers to a systematic process in which all aspects of an enterprise are deeply integrated with digital technology. Information on an enterprise's digital transformation includes multiple aspects of production and operation. Referring to Liu et al. [27], this study characterizes the degree of digitalization of enterprises by the frequency of the appearance of 76 relevant keywords in annual reports. Owing to the typical "right-skewness" feature of keyword frequency data, such data are logarithmically processed. Specifically, taking the weighted average of the degree of digitalization of listed companies among the top five customers yields the digitalization variable of the customer company ($\ln Cdigi$). Weight is the ratio of the transaction volume of the top five clients' listed enterprises to the total transaction volume of all listed client enterprises within the top five clients.

(2) Dependent variable: GTFP of state-owned suppliers ($\ln GTFP$)

This study uses an output-oriented super-efficient SBM model that considers unexpected outputs to calculate the GTFP of state-owned suppliers. The logarithm of the efficiency value obtained is taken as the GTFP of state-owned suppliers ($\ln GTFP$).

Suppose n independent individual state-owned suppliers can obtain O_1 types of expected output g ($g \in S^{O_1}$) and O_2 types of unexpected output u ($u \in S^{O_2}$) by consuming I types of factor input x ($x \in S^I$); then, the following matrices are formulated: $X = [x_1 \cdots x_n] \in S^{I*n}$, $G = [g_1 \cdots g_n] \in S^{O_1*n}$, and $U = [u_1 \cdots u_n] \in S^{O_2*n}$. Assuming that the datasets are non-negative, that is, $X \geq 0$, $G \geq 0$, and $U \geq 0$, the production possibility set of state-owned suppliers can be defined in the form of sets, meeting

$$P = \{(x, g, u) | x \geq X\lambda,\ g \leq G\lambda,\ u \geq U\lambda,\ \lambda \geq 0\} \quad (2)$$

where P is a closed set and λ is a vector of non-negative weight coefficients. The super-efficient SBM model can be obtained by solving Eq. (3).

$$\min\delta = \frac{1+\frac{1}{I}\sum_{i=1}^{I}\frac{s_i^x}{x_{ik}}}{1-\frac{1}{O_1+O_2}\left(\sum_{p=1}^{O_1}\frac{s_p^g}{g_{pk}}+\sum_{q=1}^{O_2}\frac{s_q^u}{u_{qk}}\right)}$$

$$\text{s.t.}\begin{cases} x_{ik} \geq \sum_{j=1,j\neq k}^{n}\lambda_j x_{ij} - s_i^x, i=1,\cdots,I \\ g_{pk} \leq \sum_{j=1,j\neq k}^{n}\lambda_j g_{pj} + s_p^g, p=1,\cdots,O_1 \\ u_{qk} \geq \sum_{j=1,j\neq k}^{n}\lambda_j u_{qj} - s_q^u, q=1,\cdots,O_2 \\ 1-\frac{1}{O_1+O_2}\left(\sum_{p=1}^{O_1}\frac{s_p^g}{g_{pk}}+\sum_{q=1}^{O_2}\frac{s_q^u}{u_{qk}}\right) > 0 \\ \lambda_j \geq 0, j=1,\cdots,n(j\neq k) \\ s_i^x, s_p^g, s_q^u \geq 0 \end{cases} \quad (3)$$

where k ($k = 1,\cdots,n$) is the state-owned supplier to be tested and δ is the target efficiency value. When δ is greater than or equal to 1, the state-owned supplier's GTFP is effective. The first three constraints define the relationship between the actual factor input, expected output, and unexpected output of a specific state-owned supplier and the target value; s_i^x, s_p^g, and s_q^u are the relaxation variables of the factor input, expected output, and unexpected output, respectively.

The input indicators are net fixed assets, labor, and comprehensive energy consumption. The expected output indicator is mainly business income, whereas the unexpected output indicator is pollution emissions. The unexpected output is the sum of the equivalent air and water pollution caused by an enterprise's economic behavior. It is calculated by obtaining the pollution equivalent of each pollutant according to the calculation formula and then adding them up. To avoid data bias, this study adds 1 to the total pollution equivalent and then takes the logarithm as the unexpected output of the enterprise's GTFP [8].

(3) Control variable

This study uses the following variables as control variables to improve the accuracy of the regression model: (i) size (*size*) represents the natural logarithm of the total assets of the enterprise; (ii) age (*age*) represents the natural logarithm of the duration of the existence of the enterprise plus 1; (iii) capital–labor ratio (*klratio*) represents the logarithm of the total assets divided by the number of workers; (iv) the labor productivity of enterprises (*lp*) is expressed as the logarithm of the main business income divided by the number of employees. Table 1 presents the descriptive statistics of the main variables.

4.3. Core variable

Considering data availability, this study selects data on Chinese listed state-owned enterprises from 2007 to 2022. The annual reports of listed enterprises are obtained from the official websites of the Shenzhen and Shanghai Stock Exchanges. In addition, basic information, financial indicators, governance structure, equity nature, and other enterprise data are obtained. Based on the annual reports, supplemented by social responsibility reports and websites of listed companies, data on the energy consumption and pollution emissions of enterprises are compiled. Data on industry intermediate input value, information about the top five customers, customer concentration, investment in pollution control, and environmental protection concepts of the enterprise are obtained by searching the China Stock Market & Accounting Research Database and cross-comparing the results with those from the China Research Data Service Platform database.

Using the data obtained, this study undertakes the following processing steps: (1) Listed companies classified as ST, *ST, and PT in the current year are eliminated. (2) Samples with missing key variables are removed. (3) Samples with evident recording errors, such as total assets, fixed assets, and management expenses, that are <0 are deleted. (4) The variables related to price are adjusted using a price deflator. (5) Tail trimming is conducted on continuous variables at the 1 % and 99 % percentiles to reduce the impact of extreme values on regression results, resulting in 1510 observed values. (6) The comprehensive energy consumption, which includes the use of energy such as water, electricity, and coal, is uniformly converted into standard coal units to obtain the indicator. The conversion coefficient is obtained from the official website of the Energy Bureau. (7) The pollution emission equivalent includes water and air pollution emissions, and the logarithm of the total pollution equivalent is calculated as the pollution emissions from state-owned suppliers according to Cheng et al. [8].

Table 1
Descriptive statistics of the main variables.

Variable	Observations	Mean	Std. deviation	Minimum	Maximum
ln*GTFP*	1510	−2.37	0.95	−4.84	0.46
ln*Cdigi*	1510	3.32	1.30	0.69	6.51
size	1510	22.87	1.49	19.67	26.83
age	1510	2.80	0.38	0.69	3.69
klratio	1510	2.78	4.40	0.14	74.73
lp	1510	2.21	6.02	0.07	103.18

5. Empirical results

5.1. Benchmark regression

This study adopts a multidimensional fixed effects model to conduct the benchmark regression specified in Eq. (1); Table 2 lists the regression results. Excluding fixed effects, columns (1) and (2) show the regression results with and without control variables, respectively. The results show that the digitalization coefficient of customer enterprises is positive but not significant. Columns (3), (4), and (5) report the regression results controlling for industry fixed effects; year and industry fixed effects; and year, industry, and region fixed effects, respectively. The consistent results indicate that the digitalization of customer enterprises exerts a positive impact on the GTFP of state-owned suppliers. After controlling for fixed effects, the regression results are found to be significant at the 1 % level. Thus, H1 is verified.

With the results in column (5) of Table 1 as an example, the coefficients of state-owned supplier size (*size*), labor–capital ratio (*klratio*), and labor productivity (*lp*) are significantly positive. Specifically, state-owned suppliers are forced to reform and upgrade to meet the new demands of downstream customers for digital transformation, including the greening demands from the government and downstream customers. As the scale of state-owned suppliers increases, their responsibilities in the green supply chain also increase. Therefore, state-owned suppliers actively establish environmental protection management systems to maintain their public image through green upgrades.

5.2. Robustness tests

Robustness tests are conducted by replacing the main variables and using instrumental variable methods.

(1) Replacement of the main variables. The measurement bias of variables distorts the authenticity of the estimated coefficients. This study reexamines the dependent and core explanatory variables after replacing them to overcome endogeneity. (i) Replacement of the dependent variable. This study recalculates efficiency based on the input–output-oriented super-efficiency SBM model and uses the natural logarithm as the alternative indicator, denoted as ln*GTFP*1. (ii)

Table 2
Benchmark regression results.

Variable	(1) ln*GTFP*	(2) ln*GTFP*	(3) ln*GTFP*	(4) ln*GTFP*	(5) ln*GTFP*
ln*Cdigi*	0.045	0.016	0.033**	0.041***	0.046***
	(1.63)	(0.74)	(2.06)	(2.65)	(2.99)
ln*Cdigi*1					
size		0.313***	0.319***	0.344***	0.333***
		(11.12)	(13.02)	(14.63)	(13.53)
age		−0.117	−0.148*	0.177**	0.148*
		(−1.20)	(−1.89)	(1.97)	(1.70)
klratio		0.016	0.021**	0.022**	0.022**
		(1.36)	(2.10)	(2.14)	(2.16)
lp		0.043***	0.032***	0.032**	0.031**
		(4.52)	(5.10)	(2.31)	(2.28)
_cons	−2.519***	−9.401***	−9.496***	−11.006***	−10.693***
	(−23.70)	(−14.75)	(−17.48)	(−19.43)	(−18.03)
Industry FE	No	No	Yes	Yes	Yes
Year FE	No	No	No	Yes	Yes
Region FE	No	No	No	No	Yes
N	1510	1510	1507	1507	1507
R^2	0.003	0.373	0.627	0.666	0.673

Note: The values in parentheses are *t*-test values; ***, **, and * represent significant values at the 1 %, 5 %, and 10 % levels, respectively, the same below.

Replacement of the core independent variable. This study calculates the average degree of digitalization of listed companies among the top five customers, takes the natural logarithm, and replaces the original core independent variable, denoted as ln*Cdigi*1.

According to columns (1)–(3) of Table 3, the estimated coefficients after replacing the dependent variable, independent variable, and both are positive and significant at the 5 % level. Consistent with the benchmark regression, the results indicate that the digitalization of customer enterprises significantly improves the GTFP of state-owned suppliers.

(2) Instrumental variable method. Referring to Zhang et al. [45], this study uses the number of fixed telephones per 100 individuals in each prefecture-level city in 1984 multiplied by the degree of digitalization of other enterprises in the region where the target enterprise is located as the instrumental variable (*iv*1) to further eliminate the influence of endogeneity. The second instrumental variable (*iv*2) is the 1 year-lagged Internet penetration rate of each province multiplied by the degree of digitalization of other enterprises in the region where the target enterprise is located. From the perspective of economic significance and relevance, the emergence of informatization and the development of digital technology began with the telephone dial-up access technology. By contrast, information communication, technical exchange, and the construction of digital platforms rely on the Internet. The decision-making process of enterprise digitalization is influenced by nearby enterprises; therefore, instrumental variables meet the requirements of correlation and exogeneity.

Column (4) of Table 3 shows the regression results for the instrumental variables. The *LM* statistic is 11.677, which is significant at the 1 % level, and the instrumental variable passes the unidentifiable test. The *Wald F* statistic shows that no weak instrumental variable problem exists in the regression results; the *p*-value of the *HensenJ* statistic is 0.4632, which passes the over-identification. Consistent with the results of the benchmark regression, the coefficient of customer enterprise digitalization (ln*Cdigi*) is significantly positive at the 5 % level.

Table 3
Robustness test results.

Variable	(1) ln*GTFP*1	(2) ln*GTFP*	(3) ln*GTFP*1	(4) ln*GTFP*
ln*Cdigi*	0.043***			0.553**
	(2.89)			(2.40)
ln*Cdigi*1		0.035**	0.033**	
		(2.29)	(2.23)	
size	0.281***	0.334***	0.281***	0.320***
	(11.49)	(13.52)	(11.48)	(9.39)
age	0.168**	0.150*	0.170**	0.159
	(1.98)	(1.70)	(1.98)	(1.42)
klratio	0.022**	0.022**	0.022**	0.027**
	(2.08)	(2.15)	(2.07)	(2.08)
lp	0.033**	0.031**	0.033**	0.017
	(2.42)	(2.30)	(2.43)	(1.11)
_cons	−9.686***	−10.671***	−9.667***	
	(−16.57)	(−17.84)	(−16.40)	
Industry FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes
LM				11.677***
Wald F				10.476
Hense J p-value				0.4632
N	1507	1507	1507	1507
R^2	0.660	0.671	0.658	−0.3469

5.3. Heterogeneity analysis

Various external and internal factors of the supply chain influence the information spillover effect of customer enterprise digitalization. This study conducts a heterogeneity analysis of endogenous and exogenous characteristics to further examine the spillover effects of customer enterprise digitalization on state-owned upstream suppliers. The results provide empirical evidence for further examining the underlying mechanisms.

(1) Heterogeneity of region. Owing to the differences in economic development levels and environmental regulatory intensity among different regions, a heterogeneous external development environment may have varying impacts on the green production and operational performance of enterprises. Therefore, referring to Jiang et al. [20], this study divides 30 regions in China (excluding Xizang) into eastern, central, and western regions according to the level of economic development. A regression analysis is conducted on state-owned suppliers in the three economic regions. Columns (1)–(3) of Table 4 present the results. The digitalization of customer enterprises can significantly improve the GTFP of state-owned suppliers in the eastern and western regions. However, the impact on state-owned suppliers in the central region is insignificant.

(2) Heterogeneity of factor intensity. The intensity of production factors has a significant impact on the production efficiency of enterprises, particularly in the context of digitalization and

Table 4
Heterogeneity analysis results.

Variable	Eastern	Central	Western	Labor intensive	Resource intensive	Technology intensive	High-tech	Non-high-tech	Heavy pollution	Non-heavy pollution	Key pollution control	Non-key pollution control
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
	ln*GTFP*	ln*GTFP*	ln*GTFP*	ln*GTFP*	ln*GTFP*	ln*GTFP*	ln*GTFP*	ln*GTFP*	ln*GTFP*	ln*GTFP*	ln*GTFP*	ln*GTFP*
ln*Cdigi*	0.067***	0.018	0.055**	0.003	0.047*	0.068***	0.052***	0.018	0.037*	0.059***	−0.018	0.054***
	(2.79)	(0.65)	(2.50)	(0.12)	(1.71)	(3.02)	(3.10)	(0.76)	(1.69)	(2.81)	(−0.79)	(3.28)
size	0.340***	0.364***	0.422***	0.292***	0.230***	0.436***	0.405***	0.258***	0.347***	0.330***	0.301***	0.321***
	(10.30)	(6.86)	(9.88)	(7.39)	(5.16)	(11.92)	(12.31)	(7.35)	(8.63)	(10.62)	(6.58)	(12.64)
age	0.295**	−0.183	−0.159	0.029	0.535**	0.196	0.147	0.125	0.039	0.181*	0.384*	0.137
	(2.49)	(−0.95)	(−0.94)	(0.28)	(2.37)	(1.29)	(0.99)	(1.26)	(0.28)	(1.76)	(1.80)	(1.55)
klratio	0.013	0.089**	−0.062	−0.005	0.036*	−0.040	−0.032	−0.001	0.035	0.017	0.003	0.022**
	(1.21)	(2.58)	(−1.30)	(−0.50)	(1.87)	(−1.04)	(−1.05)	(−0.11)	(1.49)	(1.41)	(0.18)	(2.08)
lp	0.037*	0.023*	0.115**	0.115***	0.033	0.053**	0.050***	0.123***	0.020	0.036*	0.188***	0.027**
	(1.84)	(1.70)	(2.49)	(3.97)	(1.25)	(2.09)	(2.61)	(4.57)	(1.21)	(1.92)	(4.17)	(2.07)
_cons	−11.199***	−10.615***	−11.955***	−9.439***	−9.544***	−13.026***	−12.309***	−8.896***	−10.926***	−10.578***	−10.957***	−10.355***
	(−13.17)	(−9.20)	(−11.63)	(−8.81)	(−9.44)	(−14.99)	(−15.92)	(−9.90)	(−12.27)	(−14.00)	(−9.13)	(−17.16)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	796	434	263	520	455	532	753	754	599	908	229	1273
R^2	0.661	0.701	0.787	0.764	0.715	0.615	0.663	0.738	0.720	0.644	0.816	0.668

greening. Hence, the pollution level of different factor-intensive enterprises also varies, resulting in different transformations and upgrading paths for state-owned suppliers. Therefore, the industry types of state-owned suppliers are divided into labor-, resource-, and technology-intensive enterprises according to production factors. Columns (4)–(6) of Table 4 present the results of the heterogeneity analysis based on intensive type. The digitalization of customer enterprises can significantly improve the GTFP of resource- and technology-intensive state-owned suppliers but has no significant impact on the GTFP of labor-intensive state-owned suppliers.

(3) Heterogeneity of high-tech level. High-tech enterprises are highly sensitive to digitalization-related elements such as information and technology. Therefore, state-owned suppliers are classified and regressed according to whether they are non-high-tech enterprises. As shown in columns (7) and (8) of Table 4, the digitalization of customer enterprises can improve the GTFP of high-tech state-owned suppliers but has no significant impact on non-high-tech and key pollution control enterprises.

(4) Heterogeneity of pollution degree. Under the goal of sustainable development, the degree of green production directly affects the green production performance of enterprises. In addition, the green requirements from cooperative enterprises and public opinion may put some pressure on polluting enterprises, leading to financing difficulties for heavily polluting state-owned suppliers and an unstable relationship between upstream and downstream suppliers. Moreover, the production efficiency of key pollution control units is affected by policies and requirements related to greening, such as pollution control and emissions. Columns (9)–(12) of Table 4 show the heterogeneity results based on the different pollution levels and whether they are key pollution control units. The results show that the digitalization of customer enterprises can significantly improve the GTFP of non-heavily polluting and non-key pollution control state-owned suppliers. Moreover, the regression coefficient of ln*Cdigi* for state-owned suppliers under key pollution control is negative but not significant.

5.4. Mechanism test

According to the theoretical analysis, the digitalization of customer enterprises may have a positive impact on the GTFP of state-owned suppliers. That is, the implicit costs of state-owned suppliers are reduced, and state-owned suppliers are forced to increase the proportion of highly educated human resources. Based on the mechanism test model, this study verifies the potential mechanism of the digitalization of customer enterprises to improve the GTFP of state-owned suppliers.

5.4.1. Mechanism test of enterprise implicit costs

Digitalization can improve the GTFP of enterprises mainly because it enhances the sharing of innovation subject information and resources within and among enterprises. In an incomplete contract environment, the new business format encourages enterprises to shift to a customer-centric paradigm. Customer information is also a competitive factor for suppliers. The digitalization of customer enterprises improves the information acquisition costs of state-owned suppliers and enhances supply chain transparency. Improving the information transparency of customer enterprises is conducive to maintaining stable supply chain relationships, reducing information verification costs and transaction risks, and further reducing the implicit costs of state-owned suppliers. State-owned suppliers can invest additional resources to improve their green production capacity and operational efficiency. This study selects management expenses and management expense rates as proxy variables for implicit and explicit costs, respectively. The objective is to test the impact of the digitalization of customer enterprises on the cost management of state-owned suppliers. The mechanism test model is as

follows:

$$exp_cost_{it} = \beta_0 + \beta_1 \ln Cdigi_{it} + \sum Controls_{it} + \{F\} + \varepsilon_{it} \quad (4)$$

$$imp_cost_{it} = \beta_0 + \beta_1 \ln Cdigi_{it} + \sum Controls_{it} + \{F\} + \varepsilon_{it} \quad (5)$$

where exp_cost_{it} and imp_cost_{it} represent explicit and implicit costs, respectively. Table 5 presents the results of the mechanism test.

The following can be observed:

(1) The coefficient of ln*Cdigi* in column (5) is −6.445, which is significantly negative at the 10 % level. This result indicates that the digitalization of customer enterprises can significantly reduce the implicit costs of non-heavily polluting state-owned suppliers. The reasons are as follows. First, the development of digital technology improves enterprises' data integration and processing capabilities, thereby reducing information redundancy. The digitalization of customer enterprises further improves the degree of information disclosure and reduces supply chain information asymmetry, including the implicit costs of information collection, verification, and processing for state-owned suppliers. Second, the technology and knowledge spillovers brought about by the digitalization of customer enterprises are beneficial for state-owned upstream suppliers to learn and imitate. Thus, their upgrade cycle is shortened, and the transformation efficiency is improved. Third, the digitalization of customer enterprises increases the demands of the consumer end, which is conducive to the precise service of state-owned suppliers to their customer enterprises. This case not only stabilizes the supply chain relationship but also reduces hidden costs caused by deviations from market demand and insufficient bargaining power. The reduction in implicit costs enables state-owned suppliers to have additional funds for organizational management, technological upgrading, and greening, resulting in an improvement in GTFP.

(2) The coefficient of ln*Cdigi* in column (4) is −0.151 but not significant. This result indicates that the digitalization of customer enterprises has no significant impact on the management expense rate of state-owned suppliers, that is, the explicit cost. Specifically, state-owned suppliers generate additional cost inputs in the process of transformation and upgrading. In particular, heavily polluting enterprises face the pressure of low-carbon transformation, which may increase rather than decrease costs. Nevertheless, this case is conducive to improving the GTFP of state-owned suppliers. Thus, H2a is verified.

5.4.2. Mechanism test of enterprise human resources

With the development of industrial synergy, state-owned suppliers need to respond to the innovation needs brought about by the digitalization of customer enterprises to stabilize customers. From the information transmission perspective, customer enterprises also require state-owned suppliers to fulfill their social responsibilities and provide products and services that meet greening requirements to avoid negative media publicity. Thus, state-owned suppliers are forced to reform their original organizational structures to adjust their production modes. As the education level of employees is closely related to their skill level, the demand of state-owned suppliers for highly skilled employees and high-end talents increases. This study selects the proportion of employees with a bachelor's degree or above among the total number of employees to measure the proportion of highly educated employees of state-owned suppliers. Furthermore, the study conducts a mechanism test based on model (6). The objective is to test how the digitalization of customer enterprises affects the GTFP of state-owned suppliers through human resources.

$$high_edu_{it} = \beta_0 + \beta_1 \ln Cdigi_{it} + \sum Controls_{it} + \{F\} + \varepsilon_{it} \quad (6)$$

where $high_edu_{it}$ represents the proportion of highly educated employees of state-owned suppliers. Table 6 presents the results of the mechanism test.

The coefficient of ln*Cdigi* in columns (1, 3), and (5) of Table 6 is significantly positive at the 10 % level. This result indicates that the digitalization of customer enterprises can significantly increase the proportion of highly educated human resources of state-owned suppliers and consequently improve their GTFP. The specific reasons are as follows:

(1) From the labor force perspective, digitalization inevitably leads to a significant adjustment in the demand structure of enterprises for labor forces engaged in simple and repeated operations and highly educated human resources with strong professionalism. Thus, the optimization and upgrading of the value chain of state-owned suppliers are promoted. The introduction of relevant technical personnel is highly conducive to enhancing the learning ability of state-owned suppliers, upgrading their production modes and digitalization, and improving GTFP.

(2) State-owned suppliers accelerate the adjustment of their production modes, moving toward intelligence, modularity, and systematization. The aim is to satisfy the needs of innovation, digitalization, and high-tech products and services brought about by the digitalization of customer enterprises. The demand for

Table 5
Mechanism analysis of enterprise's implicit cost.

Variable	Full sample		Heavily polluting		Non heavily polluting	
	(1) *imp_cost*	(2) *exp_cost*	(3) *imp_cost*	(4) *exp_cost*	(5) *imp_cost*	(6) *exp_cost*
ln*Cdigi*	24.021	−0.073	164.347	−0.020	−96.445*	−0.151
	(64.010)	(−0.43)	(130.553)	(−0.12)	(56.078)	(−0.43)
size	986.044***	−1.625***	1132.426***	−1.322***	1014.336***	−1.795**
	(168.215)	(−3.44)	(410.946)	(−4.46)	(211.119)	(−2.54)
age	−799.017***	−2.088	−1.4e+03*	−0.381	−736.656**	−2.791*
	(305.735)	(−1.64)	(820.488)	(−0.26)	(313.646)	(−1.75)
klratio	−35.484	0.072	−128.948	−0.117	−85.243***	0.133*
	(58.328)	(1.21)	(154.522)	(−0.55)	(28.659)	(1.74)
lp	−35.201	−0.238***	−360.170	−0.140*	102.078*	−0.284***
	(131.010)	(−3.24)	(349.725)	(−1.78)	(52.997)	(−2.95)
_cons	−1.9e+04***	50.735***	−1.9e+04**	38.430***	−2.0e+04***	57.018***
	(3807.350)	(4.55)	(8736.174)	(5.23)	(4448.779)	(3.33)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1507	1507	599	599	908	908
R^2	0.817	0.153	0.858	0.307	0.442	0.126

Table 6
Mechanism analysis of enterprise's human resources.

Variable	Full sample		High-tech		Non-high-tech	
	(1) *high_edu*	(2) ln*GTFP*	(3) *high_edu*	(4) ln*GTFP*	(5) *high_edu*	(6) ln*GTFP*
ln*Cdigi*	1.202**		1.153**		1.401*	
	(2.59)		(2.22)		(1.78)	
high_edu		0.010***		0.012***		0.010***
		(4.48)		(3.82)		(3.75)
size	−0.212	0.344***	−0.484	0.473***	−1.528	0.291***
	(−0.29)	(12.01)	(−0.52)	(13.60)	(−1.49)	(7.76)
age	−4.387	0.121	−0.785	−0.063	−6.594*	0.138
	(−1.61)	(1.50)	(−0.26)	(−0.49)	(−1.74)	(1.40)
klratio	1.610***	−0.021	4.407***	−0.265***	1.267**	−0.018*
	(2.67)	(−1.38)	(4.47)	(−5.76)	(2.19)	(−1.73)
lp	0.312	0.084*	−1.586**	0.195***	1.377*	0.125***
	(0.57)	(1.96)	(−2.08)	(6.14)	(1.81)	(5.11)
_cons	38.238**	−10.997***	30.266	−13.241***	74.922***	−9.889***
	(2.15)	(−15.53)	(1.41)	(−16.12)	(2.81)	(−9.61)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1058	1058	516	516	542	542
R^2	0.597	0.758	0.418	0.764	0.674	0.814

employees specializing in technology R&D is increasing, particularly for some high-tech state-owned enterprises that involve meticulous production. The particularity of production leads to a greater demand for industrial robots that can perform repetitive and high-precision work from these state-owned suppliers. In addition, external factors such as government policy support and market intelligence demand accelerate the process of replacing traditional workers with industrial robots and other equipment, further improving green production efficiency. Therefore, high-tech state-owned suppliers are more significantly affected by the digitalization of customer enterprises than non-high-tech enterprises.

(3) The introduction of highly educated human resources, particularly knowledge-based employees, by state-owned suppliers is conducive to enhancing their learning ability and amplifying the positive impact of customer enterprises' knowledge and technology spillovers. State-owned suppliers pursue technological innovation through imitation and other methods, and their GTFP is further improved. Thus, H2b is verified.

6. Further analysis

The mechanism test shows that the digitalization of customer enterprises can improve the GTFP of state-owned suppliers. That is, the implicit costs of state-owned suppliers are reduced, and their proportion of highly educated human resources increases. This section further explores the regulatory effects of environmental regulation, contractual environment, corporate environmental responsibility, and competitiveness from the two aspects of external policy environment and internal management factors. The test model for the empirical analysis is set as follows:

$$\ln GTFP_{it} = \beta_0 + \beta_1 \ln Cdigi_{it} + \beta_2 c_{ikt} + \beta_3 \ln Cdigi_{it} \times c_{ikt} + \sum Controls_{it} + \{F\} + \varepsilon_{it} \quad (7)$$

where c_{ikt} is the kth moderator variable. The coefficient and significance level of the interactive term ($\ln Cdigi_{it} \times c_{ikt}$) are the focus of this section.

6.1. Regulatory effect of external factors

6.1.1. Regulatory effect of environmental regulation

High-intensity environmental regulations can lead to short-term production efficiency losses. Enterprises increase their environmental investments to meet strict environmental regulatory requirements. To meet the requirements of strict environmental regulations, enterprises may increase their environmental protection investments, which are characterized by a large amount but a low rate of return and low economic benefits. They may also crowd out productive investments, leading to a decline in enterprises' green production efficiency. From the perspective of information transmission, the new Environmental Protection Law emphasizes the importance of public participation in environmental protection construction, thus further promoting the environmental information disclosure system. Meanwhile, the publicity of news media amplifies the effects of policy implementation. Customer enterprises may require environmentally friendly green products and services to avoid negative media publicity. However, some state-owned suppliers are unable to strictly meet environmental control standards in the short term, resulting in tension between upstream and downstream suppliers. This case is not conducive to the stability of the supply chain, where business risks increase and the GTFP of state-owned suppliers' decreases. Referring to Liu and Guan [26], this study constructs the variable *erc* to measure the intensity of environmental regulations in the region where state-owned suppliers are located.

The first column of Table 7 presents the regression results for the full sample. The interaction term ln*Cdigi* × *erc*is −2.946, which is significantly negative at the 5 % level. That is, environmental regulation weakens the positive impact of the digitalization of customer enterprises on the GTFP of state-owned suppliers and plays the role of a "speed bump" in the short term. Columns (2)–(4) and (5)–(7) show the classification regression results for the different regions and intensive types, respectively. The results indicate that high-intensity environmental constraints are detrimental to the green production efficiency of state-owned suppliers in the short term, particularly for enterprises located in the eastern region and labor-intensive enterprises. The coefficient values of the interaction term ln*Cdigi* × *erc*are −9.702 and −10.130, both of which are significantly negative at the 1 % level. Thus, H3c is verified. The reason is that as the most strictly regulated and enforced region in China, the eastern region already faces stricter environmental regulations and higher governance costs than the other regions. With the implementation of environmental regulations, the development of state-owned suppliers in the eastern region is further restricted. However, labor-intensive state-owned suppliers have the disadvantage of low resource utilization and often rely on a large amount of human capital, which leads to high technology upgrading and management costs. The "compliance cost" and production reduction loss caused by

Table 7
Regulatory effect of environmental regulation.

Variable	Full sample (1) lnGTFP	Eastern (2) lnGTFP	Central (3) lnGTFP	Western (4) lnGTFP	Labor intensive (5) lnGTFP	Resource intensive (6) lnGTFP	Technology intensive (7) lnGTFP
lnCdigi	0.063***	0.118***	0.021	0.075**	0.072*	0.040	0.094***
	(3.34)	(3.56)	(0.58)	(2.28)	(1.90)	(1.23)	(3.21)
erc	−0.531	32.113*	−9.705	5.065	33.913**	−9.826*	7.995
	(−0.12)	(1.96)	(−0.58)	(0.86)	(2.55)	(−1.85)	(0.77)
lnCdigi × erc	−2.946**	−9.702**	−1.026	−1.727	−10.130***	0.437	−4.666
	(−2.12)	(−2.38)	(−0.21)	(−0.97)	(−3.15)	(0.22)	(−1.38)
size	0.333***	0.342***	0.363***	0.426***	0.293***	0.232***	0.435***
	(13.59)	(10.42)	(6.68)	(10.15)	(7.39)	(5.15)	(11.58)
age	0.142	0.300**	−0.182	−0.186	0.034	0.522**	0.184
	(1.62)	(2.57)	(−0.96)	(−1.09)	(0.33)	(2.32)	(1.20)
klratio	0.023**	0.013	0.086**	−0.064	−0.005	0.035 *	−0.039
	(2.23)	(1.20)	(2.57)	(−1.39)	(−0.55)	(1.84)	(−1.01)
lp	0.031**	0.037*	0.023	0.116**	0.115***	0.033	0.053**
	(2.28)	(1.83)	(1.64)	(2.53)	(3.97)	(1.25)	(2.09)
_cons	−10.650***	−11.425***	−10.523***	−12.031***	−9.695***	−9.466***	−13.007***
	(−17.82)	(−13.50)	(−8.65)	(−11.81)	(−8.85)	(−9.23)	(−14.36)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	1507	796	434	263	520	455	532
R^2	0.676	0.662	0.703	0.786	0.767	0.717	0.616

environmental regulations may have a crowding out effect on productive and profitable investment, leading to the production efficiency loss of state-owned suppliers in the short term and resulting in a negative impact on current GTFP.

6.1.2. Regulatory effect of the contract environment

With the increasingly detailed division of labor in society, the production of a product may involve multiple enterprises. In an incomplete contract environment, owing to the limited rationality, opportunism, and asset specificity of human beings, enterprises that rely mainly on contracts make high investments in specific assets. Moreover, most transactions among enterprises are completed by contract rather than by market pricing decisions; thus, they bear further risks of being "ripped off." Therefore, the requirements for mastering information during the transaction process are relatively high. Referring to Nunn [32], this study measures the contract intensity of the industry to which the enterprise belongs by calculating the proportion of non-market transactions in the intermediate inputs of different industries, recorded as *zr*. The calculation formula is as follows:

$$zr_k = \sum_v \theta_{kv} R_v^{neither} \tag{8}$$

where $\theta_{kv} = \frac{\mu_{kv}}{\mu_k}$, μ_{kv} is the value of intermediate input v of industry k,$\mu_k$is the value of the total intermediate input of industry k, and θ_{kv}represents the proportion of intermediate input v in the total input of industry k. $R_v^{neither}$is the proportion of input v that is no longer traded on an exchange or at the open market price.

Given that contract intensity in this study is an industry variable, simultaneously controlling the fixed industry effect may lead to multicollinearity. Thus, the fixed industry effect is deleted, and only the region and year fixed effects are controlled for analysis. Table 8 shows the regression results of the impact of the contractual environment and digitalization of customer enterprises on the GTFP of state-owned suppliers.

Table 8
Regulatory effect of contractual environment.

Variable	Full sample (1) lnGTFP	High-tech (2) lnGTFP	Non-high-tech (3) lnGTFP	Manufacturing (4) lnGTFP	Non manufacturing (5) lnGTFP
lnCdigi	0.085**	0.094**	0.066	0.102**	0.067
	(2.44)	(2.33)	(1.27)	(2.38)	(1.03)
zr	0.012	0.117	−0.236**	0.034	−0.162
	(0.12)	(0.76)	(−2.37)	(0.21)	(−1.48)
lnCdigi × zr	0.040	0.045	0.066**	0.050	0.058*
	(1.38)	(1.08)	(2.33)	(1.16)	(1.85)
size	0.409***	0.412***	0.318***	0.391***	0.370***
	(13.70)	(13.34)	(5.90)	(11.74)	(6.30)
age	−0.040	0.117	−0.398***	0.136	−0.400**
	(−0.35)	(0.80)	(−3.06)	(0.97)	(−2.61)
klratio	−0.040*	−0.043	−0.085***	−0.023	−0.400***
	(−1.84)	(−1.49)	(−5.06)	(−0.88)	(−4.36)
lp	0.032***	0.055***	0.260***	0.041**	0.077***
	(3.43)	(2.81)	(4.30)	(2.25)	(4.04)
_cons	−11.870***	−12.299***	−9.182***	−11.954***	−10.362***
	(−16.92)	(−16.52)	(−6.29)	(−16.02)	(−6.56)
Industry FE	No	No	No	No	No
Year FE	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes
N	1141	742	398	850	291
R^2	0.568	0.628	0.707	0.578	0.720

The results show that the coefficient of ln*Cdigi* × *zr*in column (1) of Table 8 is 0.040, which fails to pass the significance test. Meanwhile, the interaction items ln*Cdigi* × *zr*in columns (3) and (5) are significantly positive. The impact of the industry contract environment on the GTFP of non-high-tech or non-manufacturing state-owned suppliers is negative, indicating that the digitalization of customer enterprises can alleviate the negative impact of the industry contract degree on the GTFP of the two types of state-owned suppliers. With the improvement in the degree of digitalization of customer enterprises, the improvement in the contract environment exerts a strong effect on the promotion of the GTFP of state-owned suppliers in the high-tech and manufacturing industries. However, the interaction term is not significant (see columns (2) and (4)). The reason for this difference is that the production behavior of state-owned suppliers in high-tech and manufacturing industries involves special technology or equipment, asset specificity is strong, and supply chain relationships are relatively stable. These enterprises generally have long-term cooperation or even fixed customer groups, with little demand for an external contract environment. Therefore, the regulatory effect of the external contract environment on the GTFP of these two types of enterprises is not significant. Non-manufacturing and non-high-tech enterprises are highly replaceable, and the "supplier–customer" relationship is vulnerable. Hence, they are likely to face the risk of being ripped off. The improvement in the degree of digitalization of customer enterprises is conducive to improving the problem of upstream and downstream information asymmetry, helping to positively regulate the GTFP of state-owned suppliers. Hence, H3a is verified.

6.2. Regulatory effect of internal factors

6.2.1. Regulatory effect of environmental responsibility

The green production of enterprises is essentially a production behavior to fulfill social responsibility, which is related not only to the green decisions of managers but also to whether a good environmental responsibility culture is established within the enterprise. Good environmental responsibility enables state-owned suppliers to choose greener production and operation modes while pursuing their own economic interests. The aim is to reduce negative external impacts on society and the environment. In this study, the environmental concept index (*ECI*) in the environmental disclosure information of enterprises is selected to measure the environmental responsibility of state-owned suppliers. Table 9 presents the results of the measurement.

From the full sample perspective, the coefficient of ln*Cdigi* × *ECI*in column (1) of Table 9 is 0.056, which is significantly positive at the 10 % level. That is, corporate environmental responsibility strengthens the role of the digitalization of customer enterprises in improving the GTFP of state-owned suppliers. Columns (2)–(6) show the regression results based on whether the enterprise is a key pollution control unit and on intensive type. The interaction items in columns (3) and (5) are significantly positive. This result indicates that for non-key pollution control units and resource-intensive state-owned suppliers with less pressure on their own pollution control, the improvement of corporate environmental responsibility can amplify the positive impact of the digitalization of customer enterprises on the green production capacity of state-owned suppliers and play the role of "accelerator." For key pollution control units, improving environmental responsibility means increasing capital investment in pollution control. Such capital investment inevitably restricts or crowds out enterprises' investments in other productive projects, which may lead to a short-term decline in the productivity of state-owned suppliers. Therefore, the interaction coefficient between environmental ideas and the digitalization of customer enterprises is not significantly negative. H3d is thus verified.

6.2.2. Regulatory effect of market competitiveness

Most traditional enterprises focus on customer groups in a certain region or industry and search for customers over a relatively fixed range for a long time, resulting in a high degree of dependence on large customers. Relying on the first few major customers may weaken enterprises' market bargaining and competitiveness, although the information collection cost for enterprises with high customer density is relatively low. Particularly under an uncertain external environment or high-risk conditions, the centralization of the supply chain is likely to cause production interruptions and serious economic losses to state-owned suppliers. Therefore, the future development direction of state-owned suppliers is to explore the possibility and potential market of cooperation with enterprises in different industries. Another direction is to improve market competitiveness and reduce dependence on key customers and market substitutability. Therefore, customer concentration is selected to measure enterprises' market competitiveness (*EMC*). Table 10 presents the results of the measurement.

The results show that the interaction term between enterprises' market competitiveness and customer enterprises' digitalization is

Table 9
Regulatory effect of environmental responsibility.

Variable	Full sample	Key pollution control	Non-key pollution control	Labor intensive	Resource intensive	Technology intensive
	(1)	(2)	(3)	(4)	(5)	(6)
	ln*GTFP*	ln*GTFP*	ln*GTFP*	ln*GTFP*	ln*GTFP*	ln*GTFP*
ln*Cdigi*	0.026	−0.002	0.026	−0.011	−0.003	0.061**
	(1.35)	(−0.04)	(1.33)	(−0.33)	(−0.09)	(2.22)
ECI	−0.116	0.130	−0.236	−0.245	−0.182	−0.042
	(−0.89)	(0.63)	(−1.59)	(−1.49)	(−0.85)	(−0.23)
ln*Cdigi* ×	0.056*	−0.021	0.087**	0.043	0.121**	0.024
ECI	(1.74)	(−0.46)	(2.39)	(0.86)	(2.32)	(0.54)
size	0.324***	0.294***	0.314***	0.304***	0.197***	0.430***
	(12.92)	(6.66)	(12.04)	(7.60)	(4.75)	(11.74)
age	0.153*	0.387*	0.142	0.019	0.547**	0.198
	(1.76)	(1.78)	(1.61)	(0.19)	(2.48)	(1.29)
klratio	0.022**	0.004	0.022**	−0.005	0.037**	−0.039
	(2.24)	(0.26)	(2.14)	(−0.56)	(2.09)	(−1.02)
lp	0.031**	0.185***	0.027**	0.116***	0.030	0.053**
	(2.28)	(4.13)	(2.09)	(3.92)	(1.19)	(2.08)
_cons	−10.453***	−10.907***	−10.127***	−9.591***	−8.761***	−12.888***
	(−17.39)	(−9.22)	(−16.40)	(−8.91)	(−8.85)	(−14.93)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1507	229	1273	520	455	532
R^2	0.674	0.815	0.670	0.765	0.729	0.615

Table 10
Regulatory effect of market competitiveness.

Variable	Full sample (1) lnGTFP	Eastern (2) lnGTFP	Central (3) lnGTFP	Western (4) lnGTFP	Manufacturing (5) lnGTFP	Non manufacturing (6) lnGTFP
lnCdigi	0.016	0.017	−0.005	0.047	0.026	−0.027
	(0.80)	(0.63)	(−0.16)	(1.28)	(1.15)	(−0.87)
EMC	−0.008*	−0.019**	−0.004	−0.006	−0.007	−0.010
	(−1.78)	(−2.20)	(−0.57)	(−0.95)	(−1.29)	(−1.56)
lnCdigi × EMC	0.002**	0.004**	0.002	0.001	0.002*	0.002
	(1.98)	(2.27)	(0.96)	(0.53)	(1.68)	(1.51)
size	0.332***	0.333***	0.366***	0.409***	0.389***	0.250***
	(13.27)	(9.79)	(6.91)	(9.26)	(12.31)	(6.46)
age	0.151*	0.301**	−0.167	−0.176	0.157	0.163
	(1.74)	(2.58)	(−0.88)	(−1.04)	(1.12)	(1.53)
klratio	0.022**	0.014	0.095***	−0.049	−0.007	0.001
	(2.20)	(1.27)	(2.82)	(−0.95)	(−0.26)	(0.13)
lp	0.031**	0.037*	0.024*	0.108**	0.033*	0.120***
	(2.29)	(1.85)	(1.74)	(2.25)	(1.81)	(4.55)
_cons	−10.548***	−10.831***	−10.668***	−11.543***	−11.884***	−8.626***
	(−17.12)	(−12.10)	(−9.30)	(−10.96)	(−15.65)	(−8.42)
Industry FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	Yes	Yes	Yes	Yes	Yes	Yes
N	1507	796	434	263	850	657
R^2	0.674	0.667	0.702	0.787	0.651	0.734

positive. Moreover, the coefficients of lnCdigi × EMC in columns (2) and (4) are significantly positive at the 10 % level. This result indicates that for state-owned suppliers located in the eastern region and belonging to the manufacturing industry, the positive regulatory effect of market competitiveness is significantly enhanced with an increase in the degree of digitalization of customer enterprises, which can "accelerate" the improvement of state-owned suppliers' GTFP. Benefiting from the advantages of high marketization, policy support, innovation, and talent ecology, state-owned suppliers in the eastern region tend to consolidate their market positions through innovation. Owing to the characteristics of standardized production, the substitutability of state-owned suppliers in the manufacturing industry is low while the competitive advantage of non-manufacturing state-owned suppliers is not pure technology or a cost advantage. In particular, for non-manufacturing state-owned suppliers that provide customized services, the gains brought about by digitization do not affect their core competitiveness. Improving market competitiveness can alleviate the pressure of the degree of substitution for state-owned suppliers in the industry. This case can also improve the bargaining power of enterprises while stabilizing the supply chain relationship and alleviate the efficiency loss caused by opportunistic behaviors such as ripping off behaviors. Therefore, the development of a diversified supply chain is conducive to improving the risk resistance of state-owned suppliers. Such development is also conducive to enhancing the competitiveness of enterprises, improving the production and operation modes, and enhancing the green operation efficiency of enterprises through the impact of technology spillover brought about by the digitalization of customer enterprises. H3b is thus verified.

7. Conclusion and policy recommendations

7.1. Conclusion and future research

Under the pressure of sustainable development goals and the increasing risks in the global supply chain, enterprises are experiencing a dual transformation of "digitalization" and "greening." In addition, the "stability maintenance" role of state-owned supplier enterprises is highlighted. However, the digitalization of state-owned suppliers is unsatisfactory because of a lack of integration of business transformation and system reform. Based on data on Chinese listed enterprises from 2007 to 2022, this study examines the impact of the digitalization of downstream enterprises on the GTFP of state-owned upstream suppliers to fully leverage the role of state-owned suppliers. This study also verifies that the digitalization of customer enterprises can improve the GTFP of state-owned suppliers by reducing the implicit costs of these suppliers, forcing the reform of their organization and management systems, and increasing their proportion of highly educated human resources. Moreover, high-intensity environmental regulations lead to the loss of production efficiency in the short term, which is manifested as a "speed bump." Particularly for enterprises located in the eastern region and belonging to labor-intensive industries with high-intensity environmental regulations, the promotional effect of the digitalization of customer enterprises on the GTFP of state-owned suppliers is significantly affected by the negative impact of environmental regulations. Additionally, improving the external contract environment helps improve the inferior position of state-owned suppliers in high-tech investments. This case allows high-tech investors to obtain additional residual claims and strengthens the positive spillover effect of the digitalization of customer enterprises on the GTFP of state-owned suppliers in high-tech and manufacturing industries. Improving internal environmental responsibility and market competitiveness helps state-owned suppliers improve their green production and business performance. Such improvement also plays an "accelerator" role in the relationship between the digitalization of customer enterprises and the improvement in the green production efficiency of state-owned suppliers. In sum, this study lays a theoretical and factual foundation for understanding how downstream "digitalization" drives the "greening" of state-owned upstream suppliers.

Future research can be conducted from the following aspects: (1) this study is based on the secondary "supplier–customer" supply chain, whereas digital technology breaks the original linear supply chain pattern. A complex supply chain network structure could be included in the model for future empirical analysis. (2) With the development of digitalization, value co-creation has become the dominant logic of commerce, and the relationship among enterprises has changed from being competitive to being cooperative. Future research could examine the impact of horizontal competition from rival enterprises on the GTFP of state-owned suppliers.

7.2. Policy recommendations

According to the research conclusion, this study offers the following policy suggestions.

First, the government should improve regional environmental regulations and strengthen guidance and policy support for the digitalization and green transformation of enterprises. It should also develop differentiated environmental law enforcement and pollution emissions standards based on the resource endowment, economic foundation, and geographical advantages of each region. Moreover, the government should understand the strengths and rhythms of policy implementation through regular and irregular monitoring. It can provide capital and talent support for transforming and upgrading enterprises through preferential policies, such as tax incentives, pollution control, energy-saving transformation, and talent subsidies. In particular, for enterprises with high clean production transformation and innovation costs, the government should implement policies that combine environmental regulations and incentives. The aim is to reduce the transformation pressure on enterprises by ensuring that their emissions meet the standards and to ultimately achieve mutual benefits and win–win results.

Second, the government should establish and improve its information disclosure systems. They should also promote the construction of a supply chain coordination platform, encourage industrial integration and supply chain information sharing, and increase the degree of supply chain information symmetry. The first point is to strengthen the market-oriented constraint mechanism and protection of intellectual property rights, encourage the disclosure of enterprise information, improve the efficiency of resource allocation, promote upstream and downstream cooperation and positive spillover of the supply chain, and provide data reference for the government to formulate regulatory policies or compensation mechanisms. The second is to build a supply chain collaboration platform to promote information sharing among financial institutions, core enterprises, government departments, and professional third-party institutions. The government can rely on digital technology to deepen information synergy; promote information transparency, safe turnover, stable production, and marketing in the industrial chain; and establish a digital supply chain ecosystem.

Third, enterprises should engage in diversified cooperation and exchange. Enterprises can make full use of the technology and talent resources of universities and scientific research institutions, promote the deep integration of high-quality talents and digitalization, and explore new businesses and new fields through industry–university research cooperation. For example, leading enterprises adopt the strategy of "learning by doing" to drive the collaborative development of various entities in the upstream, middle, and downstream through exchanges, expositions, and other similar ways. Enterprises should guide their technological innovation, transformation, and upgrading to match supply and demand and achieve the overall transformation and upgrading of the industrial chain.

CRediT authorship contribution statement

Huanhuan Jin: Writing – original draft, Formal analysis. **Nanyue Jiang:** Writing – original draft, Methodology. **Weihua Su:** Supervision. **Streimikiene Dalia:** Visualization, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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